

## **IA3 TRACEABILITY CARD TEMPLATE**

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**Team name:** LUXBYNESS

**Project title:** Modern Salon Management and Booking Website

**Client/project domain:** Beauty, Wellness, and Service Commerce

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### **1. Selected use case**

- **Use case name:** Client books a styling appointment.
- **Source evidence: client statement/meeting note/requirement ID/issue URL:**
  - *Client Statement:* "Clients need to be able to pick a specific styling service, choose their preferred stylist, select an open time slot from a calendar, and lock in the booking so we don't double-book slots."
  - *GitHub Issue URL:* <https://github.com/John-owusu-ansah/DCIT-208-Group-29/issues/12>
- **Why this use case matters to the client:** It is the primary revenue driver for the salon. Automating the booking process reduces front-desk phone overhead, prevents calendar scheduling conflicts, and gives clients a frictionless, premium onboarding experience.

### **2. Use case card**

- **Actor:** Client
- **Goal:** Successfully book a specific salon service for a chosen date, time, and stylist.
- **Trigger:** The client clicks the "Confirm Booking" button on the scheduling page.
- **Preconditions:**
  1. The client has selected at least one salon service (e.g. Braiding, Manicure).
  2. The system has pulled and displayed real-time availability for the chosen date and stylist.

- **Main success flow:**

1. The client views the service catalog and selects a service.
2. The client selects an available stylist and a preferred date from the interactive calendar UI.
3. The system displays open hourly time slots for that specific day.
4. The client selects a time slot and provides their contact details (Name, Phone number).
5. The client clicks "Confirm Booking".
6. The system validates that the selected time slot is still open.
7. The system creates a booking record with the status "Confirmed", updates the stylist's calendar, and generates a unique Booking ID.
8. The system displays a success screen with the booking summary and appointment details.

- **Alternate flow:**

- *Step 2 variation:* The client has no preference for a stylist. They choose the "Any Stylist" option. The system evaluates all available staff for that time slot and assigns the least-booked stylist automatically before moving to step 3.

- **Exception/failure flow:**

- *Step 6 failure:* Another user locks in the same time slot right before the client clicks confirm. The system detects the slot is no longer available, blocks the booking creation, refreshes the calendar view, and displays a notice:  
*"Sorry, this slot was just taken. Please select an alternative time."*

- **Postconditions:**

- A new appointment record is logged in the database linked to the client and stylist.
- The selected time slot is immediately locked out on the system calendar to prevent double-booking.
- The client lands on a confirmation summary page.

### 3. User story

As a Client, I want to select a salon service, stylist, and an open time slot online, so that I can secure my appointment instantly without needing to call the salon front desk.

### 4. Acceptance criteria

- **AC1 - Successful path:**
  - Given a client has chosen a valid service, an open date/time, and a stylist,
  - When they click "Confirm Booking",
  - Then the system should persist the appointment details to the database, mark that calendar slot as unavailable, and display a confirmation page containing their unique Booking ID.
- **AC2 - Error/exception path:**
  - Given a client attempts to complete a booking for a time slot that has been reserved by someone else while their page was open,
  - When they click "Confirm Booking",
  - Then the system must reject the transaction, prevent record insertion, and display an error message prompting them to choose a different time.
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### 5. Nonfunctional-requirement link

- **NFR category:** Consistency & Performance
- **NFR statement:** The booking engine must execute slot-availability checks and complete table writes within 1.5 seconds under a concurrent load of up to 50 users.
- **Why it matters:** As a premium salon brand, poor performance or race conditions that cause double bookings ruin the customer experience and damage the business reputation. Real-time slot locking is critical for reliability.

### 6. GitHub issue link

- **Issue title:** Book an appointment
- **Issue URL:** <https://github.com/John-owusu-ansah/DCIT-208-Group-29/issues/12>

## 7. Test idea

- Test name: Verify\_Booking\_Rejection\_On\_Schedule\_Conflict
- Input/context: Simulate two browser sessions attempting to book the exact same stylist on 2026-08-15 at 10:00 AM. User A submits a fraction of a second before User B.
- Expected result: User A receives a successful booking response. User B receives an HTTP 409 Conflict state error, no duplicate row is written to the database, and User B's interface forces a calendar reload.

## 8. Request-flow thinking

- Device/interface: Web browser optimized for mobile viewports, as most clients book salon appointments on their smartphones.
- Route/screen/endpoint:
  - Frontend Route: /book/schedule
  - Backend API Endpoint: POST /api/v1/appointments
- Data sent: A JSON object payload containing the selected service ID, the chosen stylist ID, the appointment date string, the preferred time slot string, the client's full name, and the client's phone number.
- Validation point:
  - Frontend: Check that name and phone fields are populated and match standard phone length requirements.
  - Backend: The server verifies if the database entry for the stylist ID already possesses an active appointment matching the targeted appointment date and time slot.
- Database/external service touched:
  - The appointments table inserts the new reservation row.
  - The stylists table reads availability parameters to ensure the employee is actively clocked in on that day.

- **Response:** HTTP 201 Created status code along with a JSON payload returning a success flag, the unique generated Booking ID, and the confirmed date and time string.
- **Failure point:** If the internal database server cluster drops offline or times out during processing, the transaction fails to write, throwing an HTTP 500 Internal Server Error. The system must catch this error, rollback any partial changes, and alert the client that their booking was not finalized.
- **Security/privacy concern:** Protection against Mass Assignment vulnerabilities. The backend must explicitly whitelist allowed parameters so malicious users cannot inject administrative fields or alter other users' existing appointment records within the raw network request.

## 9. AI disclosure

- **AI used?** Yes. AI tool used is Google Gemini
- **What it helped with:** Organizing the official template structure and framing the software logic (preconditions, error paths, and non-functional metrics) based directly on the LUXBYNESS team setup.
- **What I changed:** I refined the placeholders to match our core salon system design, making sure the validation points and technical data parameters reflect a realistic booking application architecture.
- **What I personally verified:** I verified the user story framework follows the correct format, ensured the acceptance criteria use explicit testable boundaries, and checked that the layout meets the mandatory assignment specifications.

## 10. Integrity declaration

I understand the use case, story, acceptance criteria and traceability links submitted here, and I can explain them orally if called upon.

TYPED NAME: SAWYERR PRINCE ELNATHAN AKILAKPA

**Traceability Matrix Row (Part F)**

| <b>Source ID/Req</b> | <b>Use Case Name</b>               | <b>User Story</b>                                                                             | <b>NFR Link</b>                                              | <b>GitHub Issue URL</b>                                                                                                                       | <b>Acceptance Criteria</b>                  | <b>Test Idea</b>                                                    | <b>Status</b>              |
|----------------------|------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------|----------------------------|
| Memo #1 / Req-12     | Client books a styling appointment | As a client, I want to select a service, stylist, and slot online so I do not have to call... | Consistency and Performance (Slot locking under 1.5 seconds) | <a href="https://github.com/John-owusu-ansah/DCIT-208-Group-29/issues/12">https://github.com/John-owusu-ansah/DCIT-208-Group-29/issues/12</a> | AC1 (Success path), AC2 (Double-book block) | Concurrent submission testing on a single slot; expect one failure. | Complete / Ready to Export |